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A Teacher-Led Hadith Study Platform — Database Design

One database, two opposite workloads,
kept apart by **privilege** – not by
convention.

2405045 · 2405052 · CSE216

The domain, and why it's a modelling problem

A **hadith** is a recorded narration. The **matn** is its text.

The **isnad** is the chain of people who transmitted it:



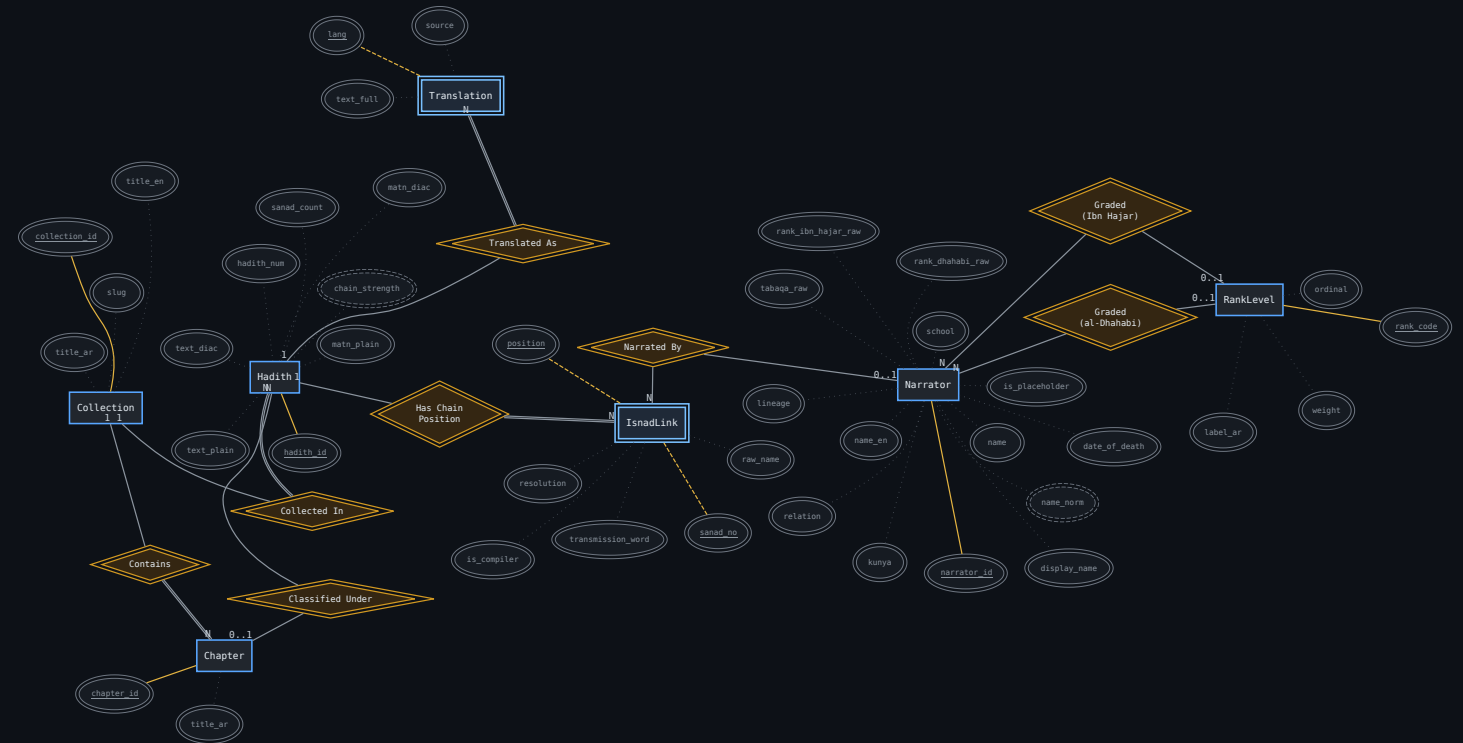
Scholars grade those narrators — **and they disagree**.

Corpus: large, shared, never edited. Study activity: small, per-user, written constantly. **Two opposite profiles, one database.**

The corpus

A chain position is
a weak entity

Identified by *(hadith, which chain, position)* —
stored explicitly, Companion first, compiler last.



The study layer

Users specialise

Disjoint: Student / Teacher / Admin. A foreign key to a *subtype* makes the rule structural — an admin cannot own a circle.

Progress is gained per
(student, hadith, assignment)

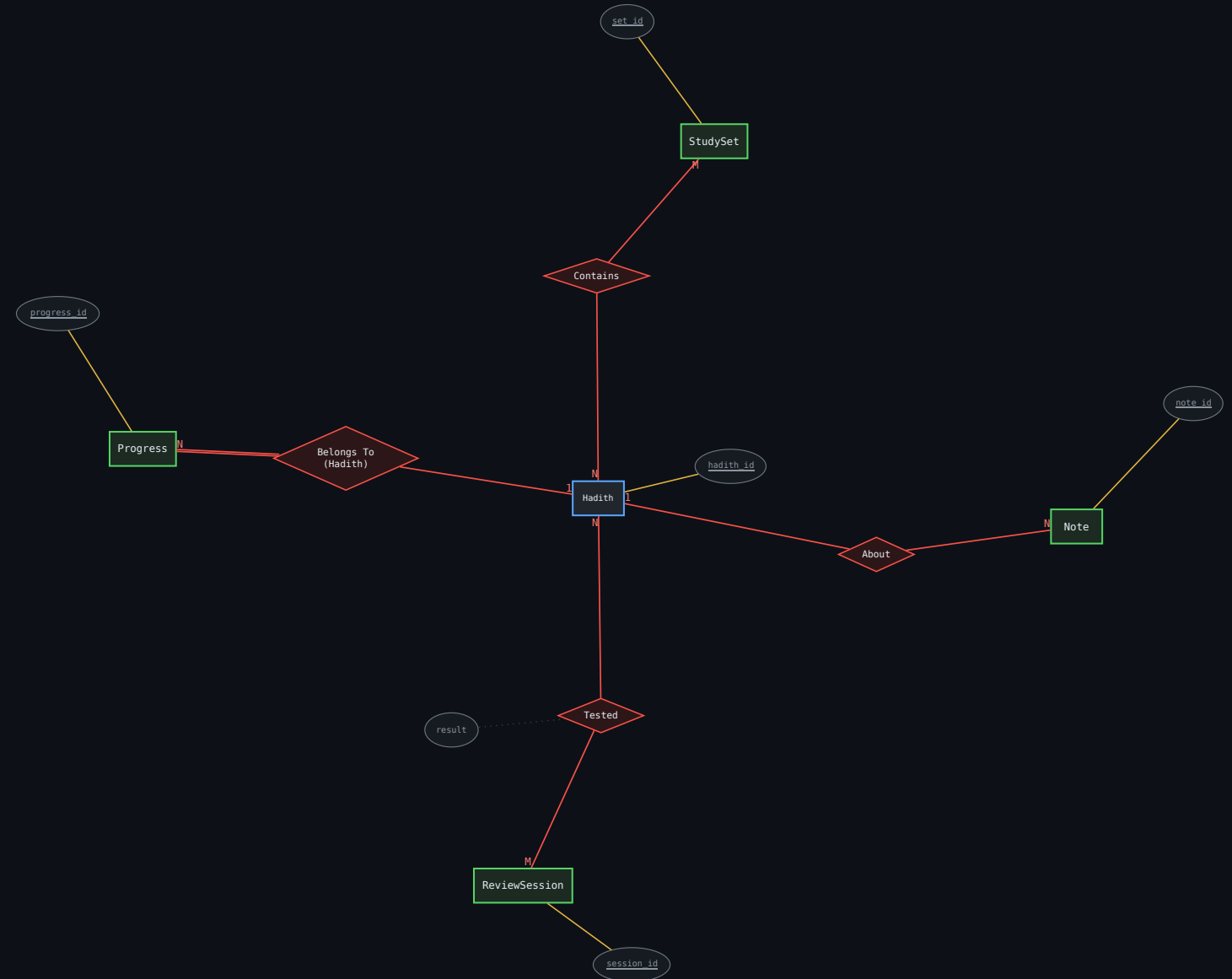
A hadith assigned twice is two obligations; prior self-study is never reset.

The boundary

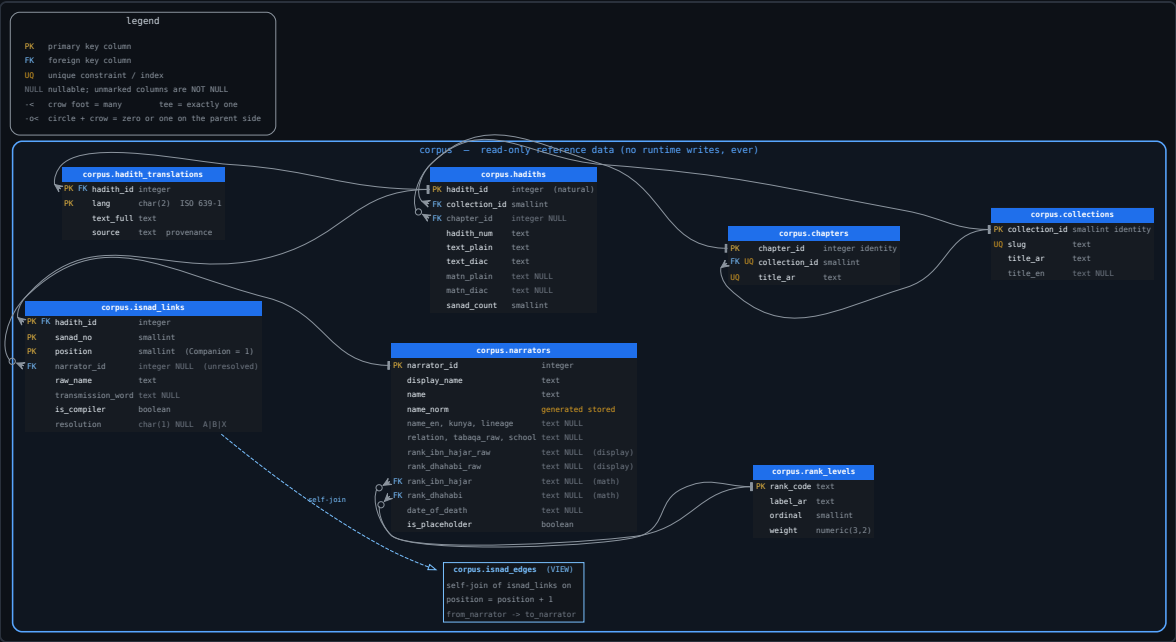
Everything the study layer knows about the corpus is here.

Four relationships, all pointing at hadith — the entire interface between the two halves.

```
REVOKE INSERT, UPDATE, DELETE  
ON corpus.* FROM <app role>;
```



It's a real schema – and what it guarantees

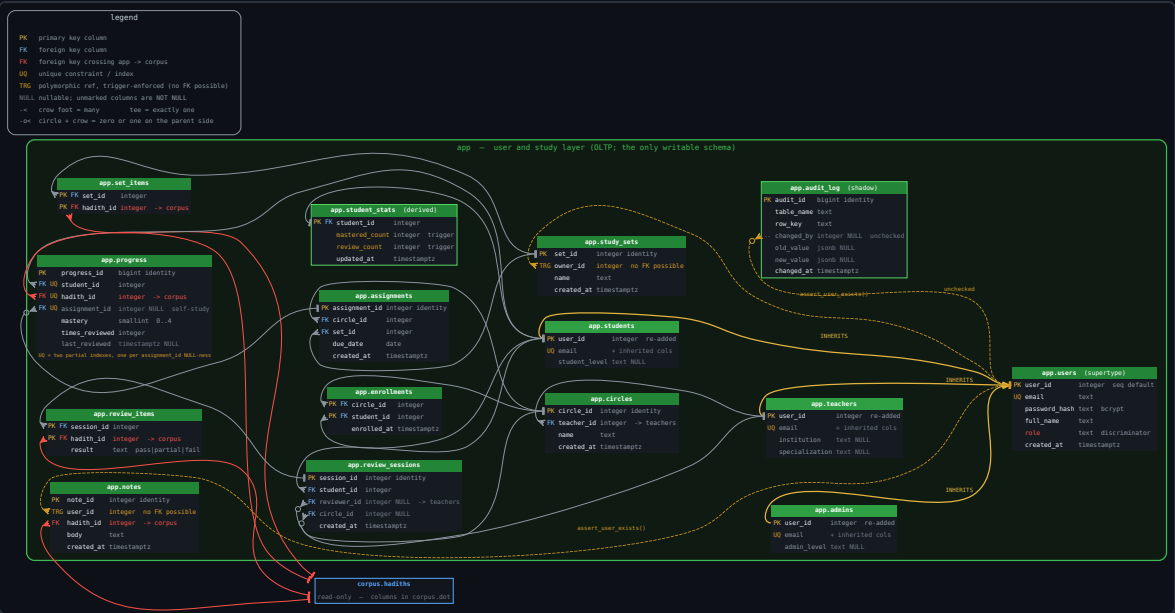


Chains are first-class

Stored positions, so traversal is aggregation — not recursion.

Disagreement preserved

Both scholars' grades kept, never collapsed into one number.



Corpus is read-only

Enforced by privilege — no application bug can undo it.